Randy Heiland

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rheiland.github.io

Education

- **Arizona State University** Tempe, AZ
  M.A. Mathematics *1992*
  Dynamical Systems. Advisor: Prof. Dieter Armbruster
  Thesis: KLTool: A Mathematical Tool to Analyze Spatiotemporal Data
- **University of Utah** Salt Lake City, UT
  M.S. Computer Science *1987*
  Computer Graphics, Computer-Aided Geometric Design. Advisor: Prof. Richard Riesenfeld
  Thesis: A Front-End User Interface to a Geometric Modeling System
- **Eastern Illinois University** Charleston, IL
  B.S. Computational Mathematics *1979*

Work Experience

- Indiana University (2003-present)
  - Intelligent Systems Engineering, 2017-present
    Research Associate
  - Center for Applied Cybersecurity Research, 2013-2017
    Senior Systems Analyst/Programmer
  - Open Systems Lab (then part of pti.iu.edu; later, became CREST) , 2007-2012
    Research Associate (~50% effort in Biocomplexity Institute)
  - UITS High Performance Applications Group, 2007
    Acting Manager
  - Scientific Data Analysis Lab, Pervasive Technology Labs, 2003-2007
    Associate Director
- Acquired Science, LLC (2005-2010)
  President. Scientific software development and consulting.
- NCSA, University of Illinois (1997-2003)
  Senior Research Scientist. Scientific visualization and data analysis.
- Pacific Northwest National Lab (1993-97). Richland, WA

Computer Scientist. Computational chemistry, image analysis, visualization.

- Los Alamos National Lab (1992). Los Alamos, NM

  Graduate Research Associate. Visualization and analysis software for the CM5 supercomputer.

- Center for Industrial Research (now SINTEF) (1985-87). Oslo, Norway

  Computer Scientist. Computer-aided geometric design software.

- Caterpillar Tractor Co. (1979-82). Peoria, IL

  Computer Scientist. Data analysis and computer-aided design.

Teaching Experience

- **Undergraduate research projects** — Bloomington
  Mentor — *2017-present*
- **Science Gateways Bootcamp** — Indianapolis
  Instructor — *April 24-28, 2017*
- **Brief Survey of Calculus (MATH 119)** — Indiana University
  Instructor — *Fall 2013*
- **Tutorial: Multi-cell, Multi-scale Modeling** — NIMBioS, UT-Knoxville
  Instructor — *May 18-21, 2011*
- **Outreach: Squeak and Scratch Modeling** — Girl Scouts of Central Indiana
  Instructor — *2007*
- **Computer Graphics I (CSC 231)** — Parkland College
  Instructor — *Fall 2001*

Notable Projects

- PhysiCell (physicell.org): An open source physics-based cell simulator.
- CTSC (trustedci.org): The NSF Cybersecurity Center of Excellence.
- CompuCell3D (compucell3d.org): Modeling environment for multi-cell behavior
- rheiland.github.io: Variety of past projects
- EMSL (emsl.pnnl.gov): Environmental Molecular Sciences Laboratory

Publications

- Google Scholar (link):

Reports and Presentations

1. R. Heiland. Cybersecurity for Science Gateways. *SGCI Bootcamp*. Indianapolis, IN. April 24-28, 2017. http://hdl.handle.net/2022/21367

2. R. Heiland and V. Welch. Center for Trustworthy Scientific Cyberinfrastructure: The NSF Cybersecurity Center of Excellence. *NSF SI2 PI Meeting: Poster session*. Arlington, VA. Feb 21-22, 2017. http://hdl.handle.net/2022/21258

3. V. Welch, *et al*. Center for Trustworthy Scientific Cyberinfrastructure - The NSF Cybersecurity Center of Excellence: Year One Report. Technical Report, Indiana University, December 2016. http://hdl.handle.net/2022/21163

4. R. Heiland, W.C. Garrison III, Y. Qiao, A.J. Lee, V. Welch. The Web's PKI: An Expository Review and Certificate Validation Cost Simulation. Technical Report, Indiana University, September 2016. http://hdl.handle.net/2022/21038

5. R. Heiland, S. Sons. Secure Software Engineering Best Practices. Presentation at the NSF Cybersecurity Summit. August 16, 2016. http://hdl.handle.net/2022/21322

6. R. Heiland, S. Koranda, V. Welch. SciGaP-CTSC Engagement Summary. Technical Report, Indiana University, May 2016. http://hdl.handle.net/2022/20926

7. R. Heiland, S. Koranda, V. Welch. SciGaP-CTSC Engagement: Final Technical Recommendations. Technical Report, Indiana University, April 2016. http://hdl.handle.net/2022/20927

8. R. Heiland, A. Adams, E. Heymann. perfSONAR-CTSC Code Review Engagement Final Report. Technical Report, Indiana University, January 2016. http://hdl.handle.net/2022/20596

9. V. Welch, *et al*. Year Three Report: Center for Trustworthy Scientific Cyberinfrastructure. Technical Report, Indiana University, Oct 2015. http://hdl.handle.net/2022/20401

10. R. Heiland and V. Welch. Analysis of authentication events and graphs using Python. *SIAM Workshop on Network Science: Poster session*. Snowbird, UT. May 2015. github.com/rheiland/authpy

11. V. Welch, *et al*. Year Two Report: Center for Trustworthy Scientific Cyberinfrastructure. Technical Report, Indiana University, Sept 2014. http://hdl.handle.net/2022/20030

12. J. Marsteller and R. Heiland. IceCube Cybersecurity Improvement Plan. Technical Report, Indiana University, 2014. http://hdl.handle.net/2022/17364

13. R. Heiland, S. Koranda, V. Welch. Globus Data Sharing: Security Assessment. Technical Report, Indiana University, 2014. http://hdl.handle.net/2022/19165

14. R. Heiland and O. Sporns. Comparing Functional Networks of the Brain: An Introductory Tutorial using Python. *BioVis 2014 Data Contest*, Boston, MA, July 2014. github.com/rheiland/biovis2014

15. R. Heiland, S. Marru, M. Pierce, V. Welch. CTSC Recommended Security Practices for Thrift Clients: Case Study - Evernote. Technical Report, Indiana University, May 2014. http://hdl.handle.net/2022/20620

16. V. Welch, *et al*. Year 1 Report: Center for Trustworthy Scientific Cyberinfrastructure. Technical Report, Indiana University, 2013. http://hdl.handle.net/2022/17205

17. R. Heiland, B. Thomas, V. Welch, C. Jackson. Toward a Research Software Security Maturity Model. *Workshop on Sustainable Software for Science*, Nov 2013. arxiv.org/abs/1309.1677

18. R. Heiland, S. Koranda, V. Welch. Pegasus-CTSC Engagement Final Report. Technical Report, Indiana University, 2013. http://hdl.handle.net/2022/15562

19. R. Heiland, J. Champlin, S. Ito, A. Litke, A. Lumsdaine, and J. Beggs. Introduction to Modeling and Computational Neuroscience using Python. *Presentation at Society for Mathematical Biology (SMB) Annual Meeting and Conference*, July 2012.

20. R. Heiland, C. Perry, B. Ream, A. Lumsdaine. Sculpture, Geometry, and Computer Science. *Presentation at SIAM Conference on Computational Science and Engineering*, Feb 2011. fperez.org/events/2011_siam_cse/siam-cse11-IndianaArc.pdf obamawhitehouse.archives.gov/blog/2010/12/10/celebrating-computer-science

21. R. Heiland. Squeak: A Free Computer Application to Enhance Math and Science Learning. *Presentation, HASTI Conference*, Indianapolis, Feb 10, 2006.

22. R. Heiland. XML for Bioinformatics. *Book Review, Briefings in Bioinformatics*, Feb 2006. https://doi.org/10.1093/bib/bbk013

23. R. Heiland, M. P. Baker, and B. D. Semeraro. A Survey of Visualization Tools for High Performance Computing. *Poster presentation at SIAM Parallel Processing for Scientific Computing*, 1999. (Abstract)

24. P. Baker, D. Bock, R. Heiland, and M. Stephens. Visualization of Damaged Structures. *CEWES MSRC PET Annual Technical Report: Year 2.* March 1998.

25. R.W. Heiland. Object-oriented parallel polygon rendering. Technical Report PNL-SA–25031; CONF-9409278–3, Pacific Northwest Lab., Richland, WA (United States), 1 September 1994.

Software Skills

- Good level: C/C++, Python(+numerous pkgs), OpenGL, Jupyter/IPython Notebooks, VTK, CMake, ParaView, MATLAB, git, LaTeX, ImageMagick, gdb, Linux, OSX
- Intermediate: Fortran, Java, R, Qt, OpenMP, ITK, HTML, Bash, Valgrind, OpenCV, VisTrails
- Basic level: Javascript, MySQL, MPI, Boost, Pegasus, CUDA, OpenCL, Django, Mathematica, Maple, Blender, Windows

Professional Society Memberships

Society for Industrial and Applied Mathematics (SIAM)

Mathematical Association of America (MAA)

Society for Mathematical Biology (SMB)

References

Available upon request.